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Computer Security

Assignment #3 (Computer Security – RSA)

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Submission Date: 28 /04/ 2026 [Tuesday] (**23:30 Hr**)

NOTE :

1. Take the **print of this sheet** and solve on it. Computer typed solution are not acceptable (**only handwritten** allowed)
2. Submissions not conforming to the given format will not be evaluated.

1. This problem illustrates a simple application of a chosen cipher text attack on the RSA cryptosystem.

Suppose TOM has an RSA key pair:

- Public key: ((n, e)) Private key: (d)

JERRY intercepts a cipher-text (C), which is an encryption of some message (M), i.e., $M = C^d \bmod n$

JERRY does not know the private key (d), but he wants to recover (M).To achieve this, JERRY performs the following steps:

1. He selects a random integer (r), such that ($0 < r < n$) and ($\gcd(r, n) = 1$).
2. He computes: $Z = r^e \bmod n$
3. He multiplies the intercepted cipher-text: $X = Z . C \bmod n$
4. He computes the modular inverse:$t = r^{-1} \bmod n$
5. JERRY then tricks TOM into decrypting (or signing) (X), and he returns:$Y = X^d \bmod n$

Q. Show that JERRY can use the value (Y), along with (t), to recover the original message (M), without knowing the private key (d).Provide a clear mathematical derivation.

Solution:

1)

$C \equiv M^e \pmod{n}$

$Z = r^e \pmod{n}$

$X = Z \cdot C \pmod{n}$

2)

$X \equiv (r^e) \cdot (M^e) \pmod{n}$

$X \equiv (r \cdot M)^e \pmod{n}$

3) Jerry tricks Tom

$Y \equiv X^d \pmod{n}$

$Y \equiv ((r \cdot M)^e)^d \pmod{n}$

$Y \equiv (r \cdot M)^{e \cdot d} \pmod{n}$

4). as $e \cdot d \equiv 1 \pmod{\phi(n)}$

$Y \equiv r \cdot M \pmod{n}$

5) Jerry computed $t = r^{-1} \pmod{n}$, he multiplies Y by t:

$Y \cdot t \equiv (r \cdot M) \cdot (r^{-1}) \pmod{n}$

$Y \cdot t \equiv M \cdot (r \cdot r^{-1}) \pmod{n}$

$Y \cdot t \equiv M \cdot 1 \pmod{n}$

$M \equiv Y \cdot t \pmod{n}$

$\boxed{11}$

Questions Based on above Attack

Q1. Given n=55, e=3, intercepted ciphertext C=14, and chosen value r = 2:

1. Compute $Z = r^e \bmod n$
2. Compute $X = Z.C \bmod n$
3. Suppose TOM decrypts X and returns Y
4. Compute $t = r^{-1} \bmod n$
5. Recover the original message M.

Solution:

$Z = r^e \pmod{n}$

1) $Z = 2^3 \pmod{55} = 8 \pmod{55} \Rightarrow 8$

2) $X = 8 \cdot 14 \pmod{55} = 112 \pmod{55} \Rightarrow 2$

3) Tom's private key = d: $n = 55 = 5 \times 11$, $\phi(n) = 40$

$e \cdot d \equiv 1 \pmod{40} \Rightarrow 3 \cdot 27 = 81 \equiv 1 \Rightarrow d = 27.$

$Y = 2^{27} \pmod{55} \Rightarrow 18$

4) $2 \cdot t \equiv 1 \pmod{55} \Rightarrow 2 \cdot 28 = 56 \equiv 1 \pmod{55} \Rightarrow 28$

5) $M = Y \cdot t \pmod{n}$

$M = 18 \cdot 28 \pmod{55} = 504 \pmod{55} \Rightarrow 9$

Write your Final answer in the table below:

Z	8
X	2
Y	18
t	28
M	9

Q2. Given:

- $n = 91$
- $Y = 65$
- $r = 4$

Find the original message M using:

- $t = r^{-1} \bmod n$

Solution:

1) $4 \cdot t \equiv 1 \pmod{91}$

$4 \cdot 23 = 92 = 91(1) + 1 \pmod{91}$

inverse $= 23$

2)

$M = Y \cdot t \pmod{n}$

$M = 65 \cdot 23 \pmod{91}$

$M = 1495 \pmod{91}$

as $1495 = 91 \cdot 16 + 39$

$M = 39$

$\boxed{IV}$

Attack-Based MCQs

1. The vulnerability exploited in this attack arises from

A. Weak key size
B. RSA's multiplicative property
C. Poor random number generation
D. Reuse of keys

2. Why must r satisfy GCD(r, n) = 1?

A. To ensure encryption is faster
B. computation of modular inverse
C. To reduce cipher-text size
D. To avoid overflow

3. This attack is successful primarily because:

A. RSA keys are public
B. decryption on attacker-controlled input
C. The modulus is small
D. The exponent is large

4. Which countermeasure prevents this attack?

A. Increasing key size
B. Using deterministic encryption
C. Using padding like OAEP
D. Reducing modulus

5. Bob computes ($X = r^e.C \bmod n$). This step:

A. Breaks RSA encryption
B. Randomizes cipher-text
C. Reduces cipher-text size
D. Reveals private key

6. If RSA were not multiplicative, this attack would:

A. Still work
B. Fail completely
C. Become slower
D. Require brute force

7. This attack shows that RSA without padding is:

A. Always secure
B. Vulnerable to chosen-cipher-text attacks
C. Only weak for small keys
D. Equivalent to symmetric encryption

Write your answers in the table given below:

Q#	1	2	3	4	5	6	8
Answer	B	B	B	C	B	B	B

V